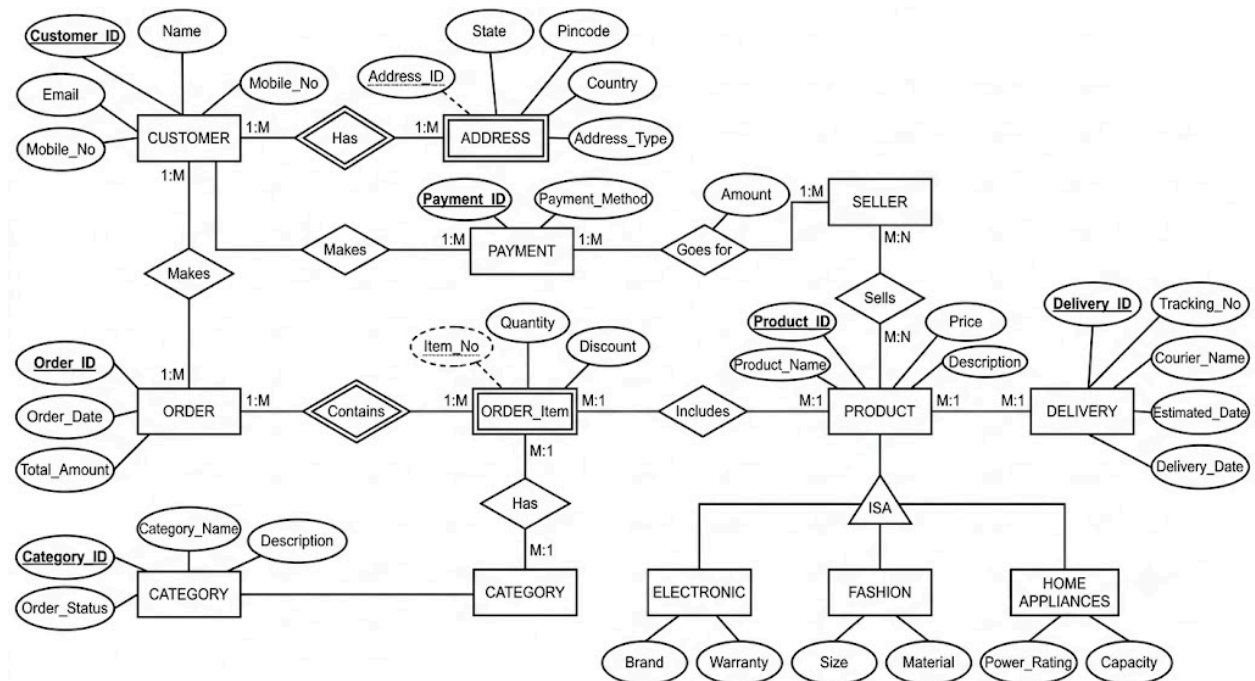


# EXPERIMENT 1

**AIM** - Design an ER diagram for an Indian e-commerce platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, Address. Specify primary keys, composite and multi-valued attributes, weak entities, specialization of Product, and participation constraints.

## ER DIAGRAM -



## OUTPUT -

- **Successful ER to Relational Mapping:** The ER diagram was successfully translated into a working relational schema. Strong entities were mapped to independent tables, while weak entities (**Address**, **Order\_Item**) were effectively implemented using composite primary keys that include the parent entity's foreign key.

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- **Specialization (ISA) Implementation:** The inheritance relationship for the **PRODUCT** entity was successfully modeled using disjoint subtypes. Sub-tables (**Electronic**, **Fashion**, **Home\_Appliances**) correctly utilized **product\_id** as both a primary key and a foreign key to link back to the parent **Product** table.
- **Prevention of Orphan Records (Insertion):** The database successfully prevented the insertion of child records that referenced non-existent parent records, throwing Error 1452 (Foreign Key Constraint Fails) when invalid **order\_id** or **product\_id** values were tested.
- **ON DELETE CASCADE Verification:** It was observed that deleting a parent record (e.g., a **Customer**) automatically and safely removed all associated child records (such as their **Orders** and **Address**) without leaving orphaned data behind.
- **ON DELETE SET NULL Verification:** Deleting a **Category** record demonstrated that the associated **Order\_Item** rows were retained in the database, but their **category\_id** foreign key was automatically updated to **NULL**, preserving the order history while removing the category link.
- **Entity and Domain Integrity Checked:** The implementation of **UNIQUE** and **NOT NULL** constraints successfully blocked the insertion of duplicate values (e.g., duplicate emails throwing Error 1062) and incomplete records (e.g., missing dates throwing Error 1364).